

Assessment Tool 2

Result and Detailed Study Report on Result Analysis

Question 1 - House Price Prediction

Algorithm: Linear Regression

Result:

House price dataset loaded. Model trained successfully. House prices were predicted with a high R^2 score, indicating good prediction performance.

Detailed Study Report (Step-by-Step Discussion):

1. Created a dataset containing Area, Bedrooms and Age.
2. Split the data into training and testing sets.
3. Trained a Linear Regression model.
4. Predicted house prices for the test data.
5. Evaluated the model using the R^2 score.
6. Conclusion: Linear Regression is suitable for predicting continuous house prices.

Question 2 - Medical Disease Prediction

Algorithm: Decision Tree Classification

Result:

Medical dataset processed. The Decision Tree classifier predicted disease status with good classification accuracy.

Detailed Study Report (Step-by-Step Discussion):

1. Prepared a dataset with Glucose, Blood Pressure and BMI.
2. Split the dataset into training and testing sets.
3. Trained a Decision Tree Classifier.
4. Predicted whether a patient has a disease.
5. Calculated classification accuracy.
6. Conclusion: Decision Trees are effective for disease classification.

Question 3 - Credit Card Fraud Detection

Algorithm: Logistic Regression

Result:

Transaction records were analysed. The Logistic Regression model classified normal and fraudulent transactions successfully.

Detailed Study Report (Step-by-Step Discussion):

1. Created a transaction dataset.
2. Separated features and target values.
3. Trained a Logistic Regression model.
4. Predicted fraud status for test records.

5. Measured model accuracy.
6. Conclusion: Logistic Regression is suitable for binary fraud detection.

Question 4 - Student Performance Prediction

Algorithm: Multiple Linear Regression

Result:

Student performance was predicted from study hours, attendance and internal marks using Multiple Linear Regression.

Detailed Study Report (Step-by-Step Discussion):

1. Prepared the student dataset.
2. Split data into training and testing sets.
3. Trained the regression model.
4. Predicted final marks.
5. Calculated the R^2 score.
6. Conclusion: Multiple Linear Regression predicts student performance effectively.

Question 5 - Loan Approval Prediction

Algorithm: Decision Tree Classification

Result:

Loan applications were classified as approved or not approved using a Decision Tree model.

Detailed Study Report (Step-by-Step Discussion):

1. Created a loan dataset.
2. Split the data into training and testing sets.
3. Trained a Decision Tree model.
4. Predicted loan approval status.
5. Calculated model accuracy.
6. Conclusion: Decision Trees are effective for loan approval prediction.